

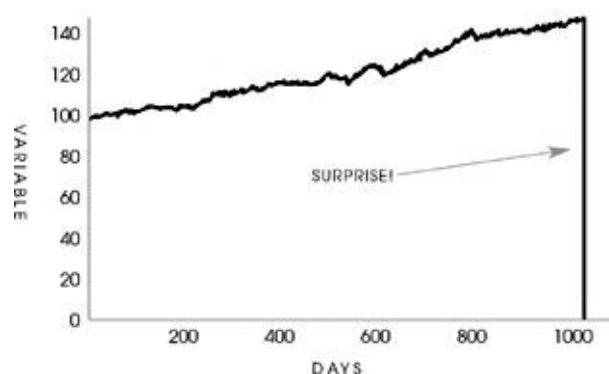
revision of belief.*

The rest of this chapter will outline the Black Swan problem in its original form: How can we know the future, given knowledge of the past; or, more generally, how can we figure out properties of the (infinite) unknown based on the (finite) known? Think of the feeding again: What can a turkey learn about what is in store for it tomorrow from the events of yesterday? A lot, perhaps, but certainly a little less than it thinks, and it is just that “little less” that may make all the difference.

The turkey problem can be generalized to any situation where *the same hand that feeds you can be the one that wrings your neck*. Consider the case of the increasingly integrated German Jews in the 1930s—or my description in [Chapter 1](#) of how the population of Lebanon got lulled into a false sense of security by the appearance of mutual friendliness and tolerance.

Let us go one step further and consider induction’s most *worrisome* aspect: learning backward. Consider that the turkey’s experience may have, rather than no value, a *negative* value. It learned from observation, as we are all advised to do (hey, after all, this is what is believed to be the scientific method). Its confidence increased as the number of friendly feedings grew, and it felt increasingly safe even though the slaughter was more and more imminent. Consider that the feeling of safety reached its maximum when the risk was at the highest! But the problem is even more general than that; it strikes at the nature of empirical knowledge itself. Something has worked in the past, until—well, it unexpectedly no longer does, and what we have learned from the past turns out to be at best irrelevant or false, at worst viciously misleading.

FIGURE 1: ONE THOUSAND AND ONE DAYS OF HISTORY



A turkey before and after Thanksgiving. The history of a process over a thousand days tells you nothing about what is to happen next. This naïve projection of the future from the past can be applied to anything.